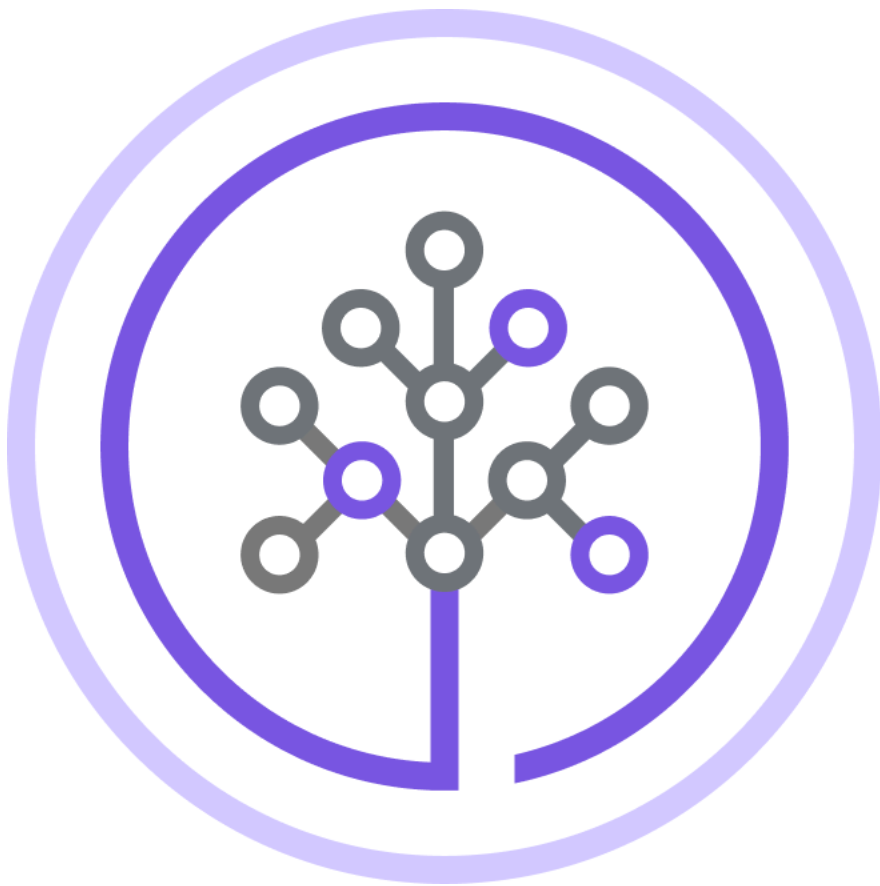


Creating Data Visualizations using ggplot

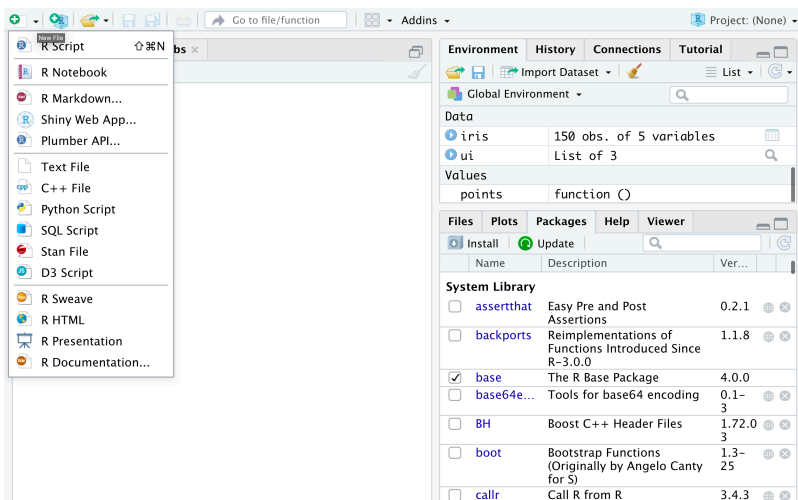


Skill Net

Objective for Exercise

We will create different data visualizations using the `ggplot` package using the inbuilt dataset in R called `mtcars`

1. Click on the + symbol on the top left and choose R Script from the menu to open a new R edit window in RStudio:



2. Read and view the first 5 rows of the Data using the following:

```
library(datasets)
#Load Data
data(mtcars)
#View first 5 rows
head(mtcars, 5)
```

3. Type this `?mtcars` to get information about the variables. This will print the information at the bottom right panel, on the Help tab

4. Copy and paste the following code to load the `ggplot` package and create a scatterplot of `disp` and `mpg`.

```
#load ggplot package
library(ggplot2)
#create a scatterplot of displacement (disp) and miles per gallon (mpg)
ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()
```

5. Use the following code to add a title.

```
#Add a title
ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement vs miles per gallon")
```

6. Use the following code to change the name of the x-axis and y-axis

```
#change axis name
ggplot(aes(x=disp,y=mpg,),data=mtcars)+geom_point()+ggtitle("displacement vs miles per gallon") + labs(x = "Displacement", y = "Miles per Gallon")
```

7. Use the following to create a boxplot of the the distribution of `mpg` for the individual Engine types vs Engine (0 = V-shaped, 1 = straight)
To do this you have to make `vs` a string or factor.

```
#make vs a factor
mtcars$vs <- as.factor(mtcars$vs)
#create boxplot of the distribution for v-shaped and straight Engine
ggplot(aes(x=vs, y=mpg), data = mtcars) + geom_boxplot()
```

8. Add color to the boxplots to help differentiate:

```
ggplot(aes(x=vs, y=mpg, fill = vs), data = mtcars) +
  geom_boxplot(alpha=0.3) +
  theme(legend.position="none")
```

9. Finally, let us create the histogram of weight `wt`.

```
ggplot(aes(x=wt),data=mtcars) + geom_histogram(binwidth=0.5)
```

This concludes this lab, we hope that you had fun!

Author(s)

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Skills Network